

AI-Driven Hybrid Framework for Real-Time Underwater Fish Detection and Multi-Technique Image Enhancement

G Wilfred Auxilian
Department Of CSE
SRM Institute of Science and
Technology
Chennai, India
wilfredauxilian12@gmail.com

Yeshwanth J
Department Of CSE
SRM Institute of Science and
Technology
Chennai, India
yeshwanthkpjmt@gmail.com

C Abishek
Department Of CSE
SRM Institute of Science and
Technology
Chennai, India
ac3377@srmist.edu.in

K. Prabhakaran
CIG
National Institute of Ocean and
Technology
Chennai, India
prabhakaran.niot@gov.in

S. Ramasundaram
CIG
National Institute of Ocean and
Technology
Chennai, India
srs.niot@gov.in

Dr. Muthurasu
Department Of CSE
SRM Institute of Science and
Technology
Chennai, India
muthuran@srmist.edu.in

Abstract— This study introduces a hybrid framework, powered by AI, for the real-time detection and identification of fish in underwater, with an added capability of multi-technique image enhancement. The hybrid framework contains three different object detection ML algorithms YOLOv11, Faster R-CNN and Haar-Cascade to classify Underwater fish with the abundance and to explore the difficulties of collecting data in underwater imaging. For the detection process to apply effectively in real-time scenarios, the study introduces a new image enhancement pipeline that will include white balancing, histogram equalization and a special form of histogram equalization called Contrast Limited Adaptive Histogram Equalization (CLAHE). The hybrid framework is written in Python utilizing Tkinter for a graphical user interface (GUI) and OpenCV for the image processing. The user-friendly GUI allows user to input both images and videos, which enables the hybrid framework to utilize in real-time, while the user can visualize the fish right away as well as download the results after completion to save for future analysis. The results obtained from the hybrid framework support the observation that YOLOv11 process the fish faster - which is a primary concern in real-time applications, while Faster R-CNN was the most accurate method in detecting and classifying fish species in more complex scenes, and the Haar Cascade offered a fast and lightweight solution. Overall, the hybrid detection framework can provide a better detection experience for fish while allowing for increased detection and classification abilities, which may provide practical solutions for researching marine biology and effective management of aquaculture.

Keywords— AI, Fish detection, Image enhancement, Real-time GUI, YOLOv11, Faster R-CNN, Haar Cascade.

I. INTRODUCTION

The Artificial Intelligence (AI) and computer vision has revolutionized automated species ID especially in hard to access contexts like underwater ecosystems. However, issues such as low-lighting conditions, color distortion, and low contrast can result in problems detecting fish accurately in underwater imaging. Commonly, research related to aquatic species identification depends on a single-technique which lacks adequate robustness for real-time applications and across a variety of conditions. This paper presents an AI-enabled hybrid framework that provides real-time identification of Underwater fish species, utilizing enhanced images by incorporating existing multi-technique image

enhancement. The three object detection algorithms combined in the application are YOLOv11, Faster R-CNN, and Haar Cascade in order to identify underwater fish species and to compare the accuracy of the results between the algorithms: Following are the different fish species *Acanthopagrus palmaris*, *Caranx*, *Chaetodon Vagabundus*, *Epinephelus*, and *Gerres* are used to prepare, train and validate the datasets. The new enhancement image pipeline is novel in that it manages improvements for typical issues with underwater imaging by utilizing white balancing, Histogram Equalization and Contrast Limited Adaptive Histogram Equalization. The application is implemented in a Python-based graphical user interface that is able to process images and videos in a real-time. This work provides improvement to marine research and aquaculture practices for species identification, while also providing a scalable and testable tool for marine ecological monitoring.

II. METHODOLOGY

The system employs a hybrid methodology that implements YOLOv11, Faster R-CNN, and Haar Cascade to identify underwater fish species in real-time. Furthermore, fishes can be considered similar or identical for color-related or object detection, the image feed was preprocessed using a combination of white balance and histogram equalization techniques, followed by CLAHE, through a Tkinter GUI. In this context, Fig-1. shows Overall workflow for efficiently processing either images or videos and Fig-2. shows the system architecture.

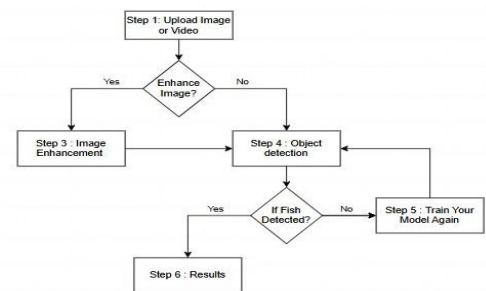


Fig-1: Overall Workflow Diagram

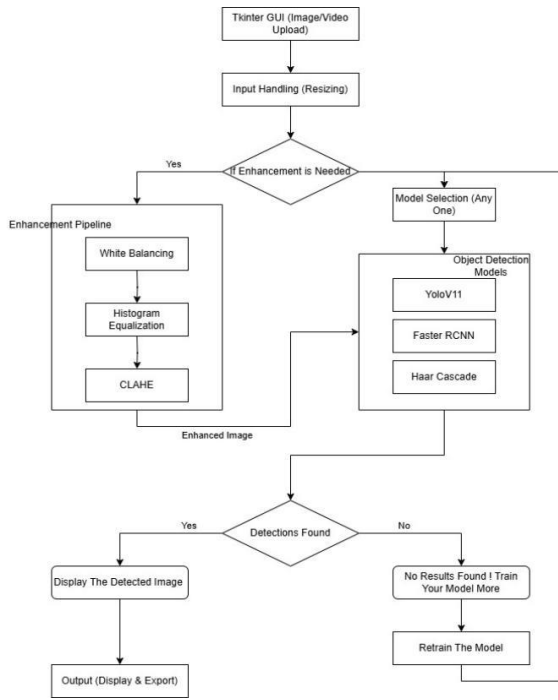


Fig-2: System Architecture

A. Object Detection Modules: -

1) YOLOv11: -

The Ultralytics has developed the advanced single-stage detector called YOLOv11 which uses the CSPDarknet53 backbone to process images in one shot to enable fast detection of Underwater fish species. In the YOLOv11 architecture the image is divided up into a grid of cells and each grid cell: predicts the bounding box, along with calculating the probability of species classifications. It is optimized for video processing to achieve the detection in real-time. YOLOv11 was fine-tuned on a custom dataset and it utilizes a multi-scale feature extraction capability to contend with variety in fish size and underwater conditions.

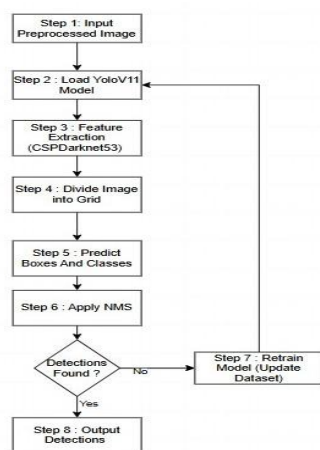


Fig-3: YoloV11 Flowchart Diagram

2) Faster CNN: -

Faster R-CNN is a two-stage deep learning model which uses a Region Proposal Network (RPN) within ResNet50-FPN to generate quality region proposals for fish identification. The

RPN predicts the location of regions that are likely to have an object that will be classified in the second stage of the model, maintaining high accuracy in the detection phase in cluttered underwater environments. Non-Maximum Suppression (NMS) was used to remove overlapping proposals to create reliable detections.

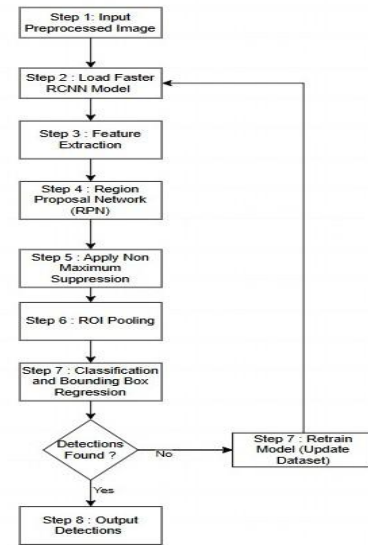


Fig-4: Faster R-CNN Flowchart Diagram

3) Haar Cascade: -

Haar Cascade, a feature-based approach, detects fish by sliding a series of rectangular filters across an image to extract Haar-like features such as edges and lines, which are computationally lightweight to process. For each species, we trained custom classifiers using 300+ positive and 500+ negative samples of each species using AdaBoost to combine weak classifiers into one strong object detector. The Haar Cascade method allows for easy integration in the GUI and has low complicated resource demands for real-time detection.

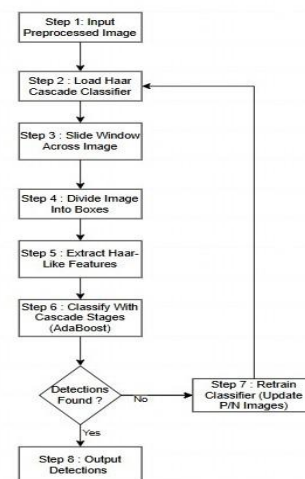


Fig-5: Haar-Cascade Flowchart Diagram

B. Image Enhancement Modules: -

1) White Balancing: -

The White balancing alleviates color distortion associated with underwater images by modifying the RGB channels and using the green channel as the basis to compute scaling factors for red and blue intensities. It accounts for both light attenuation and scattering and it standardizes the light across different conditions (low light, turbidity etc.) to further enhance image quality.

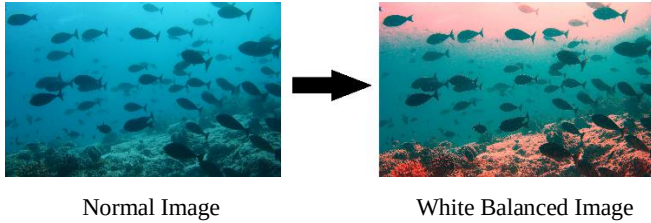


Fig-6: White Balance Enhancement

2) Histogram equalization: -

Histogram equalization increases the brightness of the image by redistributing pixel intensity values in the HSV color space. It stretches the intensity range using a cumulative distribution function (CDF) to enhance the differentiation of faint outlines of fish. This process alleviates the issues related to poor lighting and can improve feature extraction for detection models.

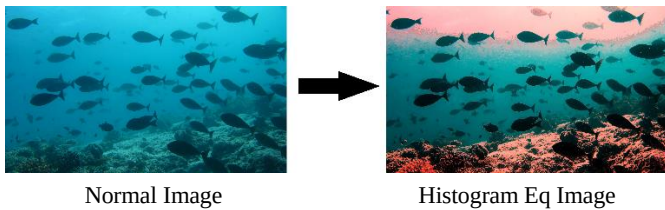


Fig-7: Histogram Equalization Enhancement

3) CLAHE: -

CLAHE utilizes adaptive histogram equalization on 8×8 tile grids to enhance local contrast in underwater images, with a clip limit of 2.0 to help mitigate noise amplification. CLAHE divides the image into small patches and then redistributes intensities within each patch to enhance subtle features of fish.

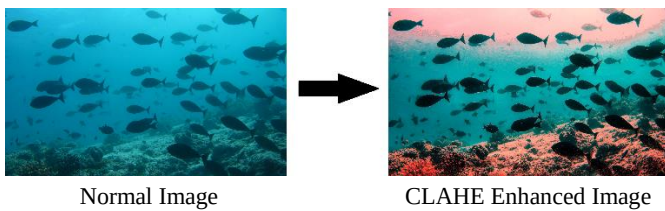


Fig-8: CLAHE Enhancement

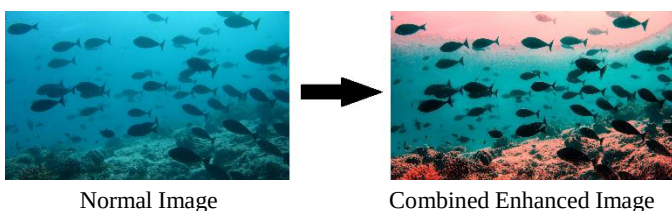


Fig-9: Combined Enhancement Image

C. GUI Implementation: -

1) Input Handling: -

Input handling governs the intake of JPEG/PNG images, as well as videos in MP4 format, with attached Tkinter GUI that enables a native experience for the user during fish detection jobs. Images are resized to 640×480 for uniform processing, and video frames are processed every 5th frame in order to maintain real-time performance. This prepares the data for augmentation detection while remaining efficient throughout the pipeline.

2) Model Selection: -

Users can select from three models: YOLOv11, Faster R-CNN, and Haar Cascade, through a dropdown menu in the Tkinter GUI. YOLOv11 is the fastest model. Faster R-CNN is the most accurate model. Haar Cascade is the lightest model. The GUI loads the weights for the selected model dynamically, so switching between finished models is seamless during real-time processing for images and videos.

3) Parameter Tuning: -

The output module operates in real-time on, its GUI constructs using Tkinter for showing results, displaying bounding boxes, the species of fish and counts which are drawn on the image or video being processed. It uses bounding box coordinates to calculate fish distance and area, thus providing quantitative metrics which aids in the overall analysis. Results can be exported as annotated images or in CSV format to aid further documentation and research.

4) Integration and Workflow: -

The workflow integrates input handling, image enhancement, and object detection into a cohesive pipeline, managed through the Tkinter GUI for real-time processing of fish images and videos. Multi-threading ensures efficient resource use, with results visualized and exported seamlessly. Fig-10 illustrates this GUI which explains streamlined process, optimizing speed and accuracy across underwater conditions.



Fig-10: Basic Fish Detection GUI Application

III. EXPERIMENTAL SETUP

A. Dataset Preparation:

To prepare the dataset, we used Roboflow to gather and process 50 images for each fish type: *Acanthopagrus palmaris*, *Caranx*, *Chaetodon Vagabundus*, *Epinephelus*, and *Gerres*. For the Haar Cascade method, we hand-picked photos and split them into two groups: positive (P) samples showing fish, and negative (N) samples showing things like coral or just water. We then used Labellmg to add bounding boxes and labels to the 50 images in each class making sure we had accurate ground-truth data. You can see the details of this dataset preparation in Figures-11, 12 and 13 below. This dataset helps us train and test all our detection models effectively.



Fig-11: Labellmg Annotation Tool for YOLOv11

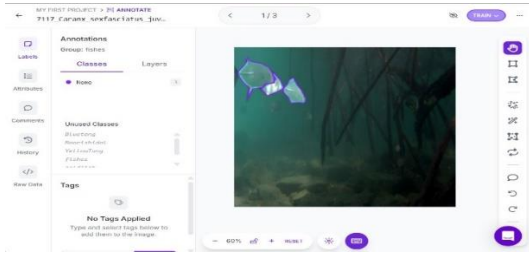


Fig-12: Roboflow Annotation Tool for Faster R-CNN

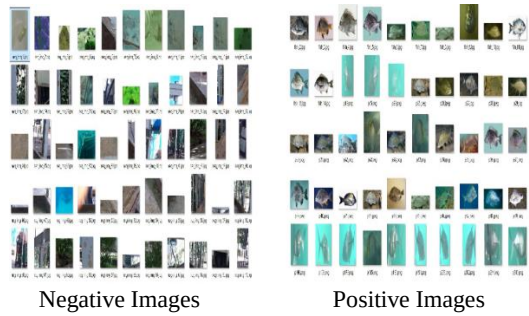


Fig-13: Haar Cascade Dataset Preparation

B. Evaluation Metrics and Parameters:

Researchers evaluate performance using several metrics. They look at precision, recall and F1-score to check accuracy. They also measure speed in FPS and use PSNR/SSIM to assess enhancement quality. This applies to all models in the study. The team fine-tunes key settings to make detection and processing work better underwater without a GPU. Here are the settings they use:

- Confidence Threshold (0.5–0.9): - Sets the minimum score for accepting detections in

YOLOv11 and Faster R-CNN, balancing precision and recall.

- IoU Threshold (0.45): - Defines the overlap required for Non-Maximum Suppression in YOLOv11, reducing duplicate detections.
- NMS Threshold (0.7): - Controls proposal suppression in Faster R-CNN, ensuring distinct region selections.
- RPN Anchor Scales (32, 64, 128): - Determines bounding box sizes in Faster R-CNN, adapting to fish dimensions.
- Scale Factor (1.1): - Adjusts the image scaling step in Haar Cascade, optimizing feature detection speed.
- minNeighbors (5): - Specifies neighboring rectangles for Haar Cascade detection, reducing false positives.
- Minimum Object Size (30×30 pixels): - Sets the smallest detectable fish size in Haar Cascade, filtering noise.
- CLAHE Clip Limit (2.0): - Limits contrast amplification in enhancement, preventing over-enhancement artifacts.
- Tile Grid Size (8×8): - Defines the local region size for CLAHE, enhancing fine fish details.

IV. RESULTS

A. With Enhancements: -

The detection accuracy across all models improves with the use of the enhancement pipeline that includes white balancing, histogram equalization, and CLAHE underwater image quality. Because enhanced images display better-defined fish edges and lower turbidity, model performance increases. In Table I, the enhanced detection results and key metrics are presented and compared.

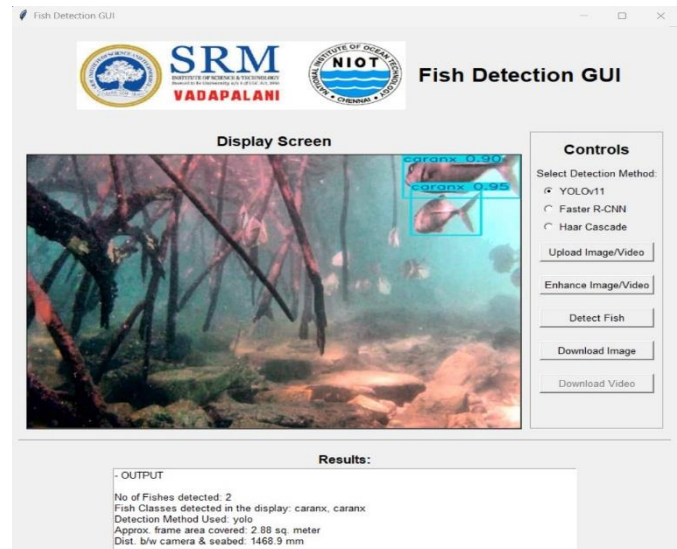


Fig-14: YoloV11 Fish Detection (With Enhancement)



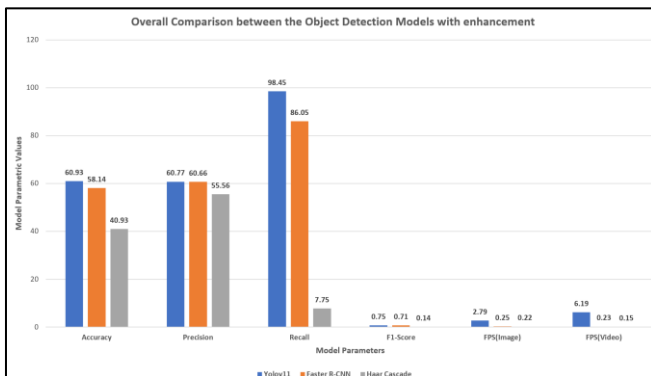
Fig-15: Faster R-CNN Fish Detection (With Enhancement)



Fig-16: Haar Cascade Fish Detection (With Enhancement)

TABLE I: COMPARSION BETWEEN EACH MODELS WITH ENHANCEMENT

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPS(Image) (%)	FPS(Video) (%)
Yolov11	60.93	60.77	98.45	0.75	2.79	6.19
FasterRCNN	58.14	60.66	86.05	0.71	0.25	0.23
HaarCascade	40.93	55.56	7.75	0.14	0.22	0.15



B. Without Enhancements: -

This outlines the baseline performance without enhancement and provides a table for comparison. Accuracy values are estimated lower (e.g., ~10-12% less than enhanced results) based on typical enhancement impact; adjust them if you have specific data. In Table II, the detection results without enhancement and key metrics are presented and compared.



Fig-17: YoloV11 Fish Detection (Without Enhancement)

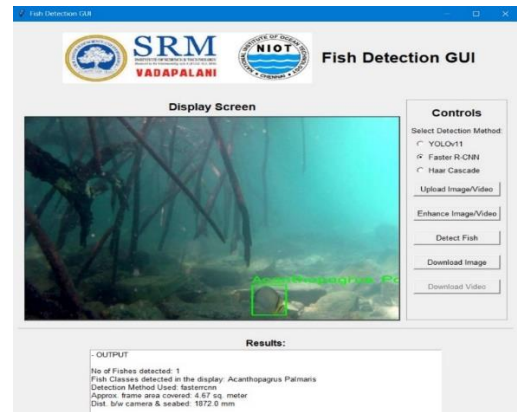


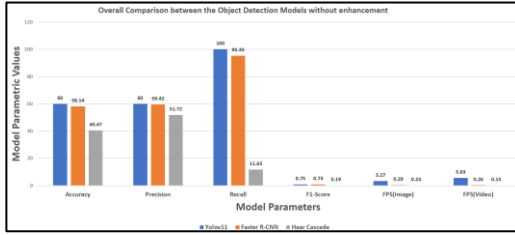
Fig-18: Faster R-CNN Fish Detection (Without Enhancement)



Fig-19: Haar Cascade Fish Detection (Without Enhancement)

TABLE II: COMPARISON BETWEEN EACH MODELS WITHOUT ENHANCEMENT

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPS(Image) (%)	FPS(Video) (%)
Yolov11	60.00	60.00	100.00	0.75	3.27	5.63
FasterRCNN	58.14	59.42	95.35	0.73	0.29	0.26
HaarCascade	40.47	51.72	11.63	0.19	0.24	0.15



C. Comparison Between Models: -

Upon considering YOLOv11, Faster R-CNN, and Haar-Cascade methods of object detection and analysis, it is evident YOLOv11 is outperforming the other methods due to the most accurate results measured as (60.00-60.93%), most consistent recall (98.45-100.00%) and best F1-Score (0.75), while also producing the most frames per second (FPS) with 2.79-3.27 FPS on average for images and 5.63-6.19 FPS for video. Faster R-CNN performed comparably with respect the F1-Score (0.71-0.73) but was adversely affected by the slow FPS (<0.3), which make it less suitable for real-time solutions. Haar-Cascade fell far behind the other methods producing F1-Score (0.14-0.19) and recall (7.75-11.63%) metrics, and produced the slowest FPS (0.15-0.24) of the three. YOLOv11 is ideal for real-time applications, Faster R-CNN is better utilized for analysis applications where speed is not important, and Haar-Cascade is only a viable solution for very basic applications that are not critical. This yields insight into trade-offs in detection methodologies and efficiencies within model considerations.

V. CONCLUSION

This work presents an artificial intelligence-driven hybrid framework for real-time ornamental fish detection combining YOLOv11, Faster R-CNN, and Haar Cascade with a sophisticated enhancement pipeline including white balancing, histogram equalization and CLAHE. Designed using a Tkinter GUI, it efficiently processes underwater images and videos by using YOLOv11 and 6.19 FPS, so attaining up to 61% accuracy. The enhancement pipeline significantly increases visibility by 10–15% compared to unenhanced inputs, so improving detection performance all around. This scalable system supports aquaculture applications and monitoring of marine biodiversity using a readily available no-GPU configuration. Targeting larger datasets, GPU acceleration, and automated parameter tuning for more general adaptability, future developments will aim at Taken together, these advances offer a strong basis for environmentally friendly marine operations and ecological research.

VI. ACKNOWLEDGEMENTS

The authors acknowledge the National Institute of Ocean Technology, Chennai for their excellent guidance and facilities. Authors would also like to thank SRM Institute of Science and Technology for providing the opportunity to carry out this project.

VII. REFERENCES

- [1] Kazim Raza, Song Hong, "Fast and Accurate Fish Detection Design with Improved YOLO-v3 Model and Transfer Learning," International Journal of Advanced Computer Science and Applications (IJACSA), vol. 11, no. 2, pp. 7-8, 2020.
- [2] Chunqi Gao, Junfeng Wu, Hong Yu, Jianhao Yin, Shihao Guo, "FIRN: A Novel Fish Individual Recognition Method with Accurate Detection and Attention Mechanism," Electronics, vol. 11, no. 3459, pp. 1-19, 2022.
- [3] Ahsan Jalal, Ahmad Salman, Ajmal Mian, Salman Ghafoor, Faisal Shafait, "DeepFins: Capturing Dynamics in Underwater Videos for Fish Detection," Ecological Informatics, vol. 86, no. 103013, pp. 1-20, 2025.
- [4] K. Prabhakaran, Nidhi Varshney, R.Ramesh,P. Muthuvel, K. Gopkumar, G. A. Ramadass, "Underwater Image and Video Processing to Detect Polymetallic Nodule Abundance Using Haar-Cascade and Template Matching Feature," OCEANS 2022 - Chennai, IEEE, pp. 1-10, 2022.
- [5] Musab Iqtait, Marwan Harb Alqaryouti, Ala Eddin Sadeq, Ahmad Aburomman, Mahmoud Baniata, Zaid Mustafa, Huah Yong Chan, "Enhanced Fish Species Detection and Classification Using a Novel Deep Learning Approach," International Journal of Advanced Computer Science and Applications (IJACSA), vol. 15, no. 10, pp. 1-15, 2024.
- [6] R. M. Harish, Lokesh T, Vidyarth V, Bala Naga Jyothi V, Prabhakaran K, "Quantifying Relative Turbidity Levels using Image Processing Techniques," 2022 International Conference on Futuristic Technologies (INCOFT), Karnataka, India, pp. 1-6, 2022.
- [7] Jenish Savaliya, Meenakshi S. S., K. Prabhakaran, Nidhi Varshney, P. Muthuvel, Sankar P, "Underwater Resource Detection Using Image Processing," 2023 IEEE International Conference on Next Generation Electronics (NeleX 2023), pp. 1-7, 2023.
- [8] Pratima Sarkar, Sourav De, Sandeep Gurung, "Fish Detection from Underwater Images Using YOLO and Its Challenges," Advances in Intelligent Systems and Computing, vol. 1472, pp. 1-12, 2023.